

Gioia Zheng

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Profile

B.Sc. student in Applied Computer Science and Artificial Intelligence at Sapienza University of Rome. Building reproducible retrieval and evaluation systems for large-scale QA pipelines. Interested in research-engineering problems at the intersection of information retrieval, evaluation methodology, and AI infrastructure.

Education

Sapienza University of Rome

Sep 2022 -- Present

B.Sc. in Applied Computer Science and Artificial Intelligence

Projects

MS MARCO Generative QA System

github.com/GioiaZheng/msmarco-genqa

Python --- Retrieval, Reranking, Generation, Evaluation

- Built an end-to-end retrieve -> rerank -> generate pipeline on MS MARCO (6,980 queries, 8.8M passages), improving Token-F1 from 0.197 to 0.368 on a frozen T5-small generator.
- Implemented sparse BM25 retrieval, dense retrieval with SBERT/FAISS, cross-encoder reranking, and T5 generation.
- Developed reproducibility infrastructure including experiment manifests, CI workflows, and automated evaluation pipelines.
- Investigated disagreement between surface-form metrics and NLI-based grounding signals through regression and failure analysis.

Handwritten OCR Pipeline

github.com/GioiaZheng/handwritten-ocr-system

PyTorch --- CRNN, BiLSTM, CTC

- Implemented a handwritten text recognition system using residual CNN layers, BiLSTM sequence modeling, and CTC decoding.
- Built training and evaluation workflows with CER/WER tracking, TensorBoard logging, and preprocessing pipelines.

Chat Service

github.com/GioiaZheng/go-chat-system

Go Backend + Vue Frontend + SQLite

- Developed a REST-based chat service with SQLite persistence and a Vue frontend bundled into a single deployable binary.
- Integrated CI validation including testing, OpenAPI verification, formatting, and frontend build checks.

Experience

Web Development Intern

Mar 2026 -- Present

Idearia --- Rome, Italy

- Developed and maintained PHP / WordPress backend systems for production websites.
- Built automation workflows using n8n to connect forms, content pipelines, and external services.

NLP Algorithm Engineer Intern

Feb 2026 -- May 2026

Microsoft --- Natural Language Computing (Remote)

- Contributed to retrieval-augmented QA experimentation and evaluation on the MS MARCO benchmark.
- Built and evaluated retrieval, reranking, generation, and analysis components over a 6,980-query benchmark using BM25, FAISS, sentence-transformers, and T5.
- Conducted reproducibility, grounding, and failure-analysis studies for RAG pipelines.

Skills

Programming

Python, Go, SQL, JavaScript

Research and AI

Information Retrieval, RAG Evaluation, Reproducibility, PyTorch, FAISS, BM25, Sentence Transformers, Cross-Encoder Reranking

Systems

Linux, Docker, GitHub Actions, OpenAPI, SQLite

Languages

English (Professional), Italian (Bilingual), Chinese (Native), German (Beginner)